SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 08

Exercise 5: Mismatched source and destination, and Buffered Send

IT24610328

On instance with 2 vCPUs (t3.micro)

Nano - creating the mismatched source code (receive source changed from 1 to 2)

```
  GNU nano 6.2                                                    messages_mismatch.cc
#include <cstdio>
#include <cstdlib>
#include <mpi.h>
int main(void)
{
    int rank;
    MPI_Status status;
    MPI_Init(NULL, NULL);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    char name[30];
    int len;
    MPI_Get_processor_name( name, &len );
    int x, y;
    x = 30;
    if (rank == 1) {
      printf("SEnding message to computer 3 from computer 1\n");
      MPI_Ssend(&x, 1, MPI_INT, 3, 0, MPI_COMM_WORLD);
    }
    else if (rank == 3) {
      // IT24610328 - Ex5 - Changed receive source from 1 to 2
      MPI_Recv(&y, 1, MPI_INT, 2, 0, MPI_COMM_WORLD, &status);
      printf("in computer 3 the value of y is %d\n", y);
    }
    else
      printf("Just a normal process From rank %d machine %s\n", rank, name);
    MPI_Finalize();
}
```

Output from running the mismatched source code (stopped by timeout after 10 seconds)

```
[ec2-user@ip-172-31-17-201 ~]$ timeout 10 mpirun --oversubscribe -np 4 ./messages_mismatch
Just a normal process From rank 0 machine ip-172-31-17-201.ec2.internal
Just a normal process From rank 2 machine ip-172-31-17-201.ec2.internal
SEnding message to computer 3 from computer 1
```

What happens

The program hangs. Rank 1 sends to rank 3 with MPI_Ssend, but rank 3 is waiting for a message from rank 2, which never sends one. Neither side can finish, so the run only ends when timeout stops it.

Output from running messages1 with 3 processes (destination rank 3 does not exist)

```
[ec2-user@ip-172-31-17-201 ~]$ timeout 10 mpirun --oversubscribe -np 3 ./messages1
Just a normal process From rank 0 machine ip-172-31-17-201.ec2.internal
Just a normal process From rank 2 machine ip-172-31-17-201.ec2.internal
SEnding message to computer 3 from computer 1
[ip-172-31-17-201:35086] *** An error occurred in MPI_Ssend
[ip-172-31-17-201:35086] *** reported by process [233504769,1]
[ip-172-31-17-201:35086] *** on communicator MPI_COMM_WORLD
[ip-172-31-17-201:35086] *** MPI_ERR_RANK: invalid rank
[ip-172-31-17-201:35086] *** MPI_ERRORS_ARE_FATAL (processes in this communicator will now abort,
[ip-172-31-17-201:35086] ***    and potentially your MPI job)
```

MPI aborts with MPI_ERR_RANK: invalid rank.

Nano - creating the message2 code with Buffered Send

GNU nano 6.2 message2_bsend.cc

```
#include <cstdio>
#include <cstdlib>
#include <mpi.h>
int main(void)
{
    int rank;
    MPI_Status status;
    MPI_Init(NULL, NULL);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    char name[30];
    int len;
    MPI_Get_processor_name( name, &len );
    int x[10], y[10];
    if (rank == 1) {
      for (int r =0;r <10; r++)
         x[r] = 10*r;

      printf("SEnding message to computer 3 from computer 1\n");
      // IT24610328 - Ex5 - Added buffer attach for Bsend
      int bufsize = 10 * sizeof(int) + MPI_BSEND_OVERHEAD;
      char *buf = (char *)malloc(bufsize);
      MPI_Buffer_attach(buf, bufsize);
      // IT24610328 - Ex5 - Changed MPI_Ssend to MPI_Bsend
      MPI_Bsend(x, 10, MPI_INT, 3, 0, MPI_COMM_WORLD);
      // IT24610328 - Ex5 - Added buffer detach and free
      MPI_Buffer_detach(&buf, &bufsize);
      free(buf);
    }
    else if (rank == 3) {
      MPI_Recv(y, 10, MPI_INT, 1, 0, MPI_COMM_WORLD, &status);
      printf("in computer 3 the value of y is printed\n");
      for (int r=0;r<10;r++)
         printf(" %d ",y[r]);
    }
    else
      printf("Just a normal process From rank %d machine %s\n", rank, name);
    MPI_Finalize();
}
```

Output from running the message2 code with Buffered Send (4 processes)

